A counterexample to Bu–Gu–Jaffe's expected mixed-state inverse sumset estimate

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Abstract

We refute the mixed-state inverse sumset estimate expected in Remark 45, equation (58), of Bu, Gu and Jaffe, *Quantum Ruzsa Divergence to Quantify Magic* [1, arXiv:2401.14385v3]. A computationally diagonal equal mixture of two basis states in dimension seven has quantum-doubling constant $\sqrt{2}$ and relative entropy $\log(7/2)$ from its mean state, while the proposed bound is at most $\frac{1}{2}\log 7$. The contradiction uses exact inequalities and persists for every tensor power. For the same family, we verify that the weaker estimate (59) has a positive denominator and holds strictly, and give the equality case of the companion entropy-growth statement for computationally diagonal states.

1 The expected estimate and its conventions

Bu, Gu and Jaffe extend entropic inverse sumset theory to discrete quantum convolution [1]. Their Remark 45 expects a mixed-state extension of the pure-state bound in Theorem 44. We show that its displayed estimate (58) fails already for a diagonal state, so the obstruction is classical rather than a consequence of noncommutativity.

Let d be an odd prime and $s, t \in \mathbb{F}_d \setminus \{0\}$ satisfy $s^2 + t^2 = 1$. On computational basis vectors, the unitary and convolution are

$$U_{s,t}|i,j\rangle = |si + tj, -ti + sj\rangle, \qquad \rho \boxtimes_{s,t} \sigma = \operatorname{Tr}_2\big[U_{s,t}(\rho \otimes \sigma)U_{s,t}^*\big]. \tag{1}$$

For n qudits, this map acts coordinatewise. All logarithms are natural; $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$, with $0 \log 0 = 0$. We use $D(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$ when $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, and $+\infty$ otherwise. We use the source's characteristic function $\Xi_\rho(p,q) = \operatorname{Tr}(\rho w(-p,-q))$, where $Z|j\rangle = \omega^j|j\rangle$, $X|j\rangle = |j+1\rangle$, $\omega = e^{2\pi i/d}$ and $w(p,q) = \omega^{-pq/2}Z^pX^q$; division by two in this exponent means multiplication by 2^{-1} in $\mathbb{F}_d$. The mean state $\mathcal{M}(\rho)$ retains precisely the characteristic coefficients of modulus one [1, Definitions 6–8 and 10]. Write

$$R(\rho) = \operatorname{rank}\mathcal{M}(\rho), \qquad \lambda = (1 - \mathrm{MG}(\rho))^2,$$
$$\delta_q[\rho] = \exp\big(S(\rho \boxtimes \rho) - S(\rho)\big), \qquad \mathrm{MG}(\rho) = 1 - \max_{0<|\Xi_\rho(x)|<1} |\Xi_\rho(x)|,$$

with the source's convention $\mathrm{MG} = 0$ when the maximizing set is empty. Its expected estimate, when $1 < \delta_q[\rho] \le C$, is

$$D(\rho\|\mathcal{M}(\rho)) \;\le\; \frac{\log R(\rho)}{\log R(\rho) - \log\big[1 + \lambda(R(\rho)\operatorname{Tr}\rho^2 - 1)\big]}\log C. \tag{BGJ 58}$$

Our examples satisfy strict positivity of every scalar logarithm argument in the displayed estimates and of their denominators, as well as the source's zero-mean condition: $\Xi_{\mathcal{M}(\rho)}$ takes values in $\{0,1\}$.

The mean-state framework was developed in [3, 2]; the classical entropic inverse problem appears in Tao [4]. Recent work of Goodarzi develops submodular entropy bounds for compatible quantum convolutions [5, Theorem 4.1]. The question here is the specific quantitative mixed-state estimate (BGJ 58). The single-qudit witness below is the construction in the trureturing source [6]. We give its elementary exposition and derive the tensor-product and equation-(59) consequences below.

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2 An exact counterexample

Theorem 1. *Take $d = 7$, $s = t = 2$ and $\rho = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)$. Then* (BGJ 58) *is false for $C = \sqrt{2}$. In fact, its right-hand side is at most $\frac{1}{2}\log 7$ even if the actual λ is replaced by any number in $[0, 1]$, whereas $D(\rho\|\mathcal{M}(\rho)) = \log(7/2) > \frac{1}{2}\log 7$.*

Proof. The parameters are admissible since $2^2 + 2^2 = 8 = 1$ in $\mathbb{F}_7$. For a diagonal state, (1) is the law of $2I + 2J$ modulo seven, where I, J are independent and uniform on $\{0, 1\}$. Thus

$$\rho \boxtimes \rho = \tfrac{1}{4}|0\rangle\langle 0| + \tfrac{1}{2}|2\rangle\langle 2| + \tfrac{1}{4}|4\rangle\langle 4|. \tag{2}$$

Consequently

$$S(\rho) = \log 2, \qquad S(\rho \boxtimes \rho) = \tfrac{3}{2}\log 2, \qquad \delta_q[\rho] = \sqrt{2} > 1, \qquad \operatorname{Tr}\rho^2 = \tfrac{1}{2}. \tag{3}$$

If $q \neq 0$, $w(-p, -q)$ has zero diagonal and hence $\Xi_\rho(p, q) = 0$. On the clock axis,

$$\Xi_\rho(p, 0) = \frac{1 + \omega^{-p}}{2}, \qquad |\Xi_\rho(p, 0)| = \left|\cos\frac{\pi p}{7}\right|. \tag{4}$$

The modulus is one only for $p = 0$. Characteristic-function inversion therefore gives

$$\mathcal{M}(\rho) = I_7/7, \qquad R(\rho) = 7, \qquad D(\rho\|\mathcal{M}(\rho)) = \log 7 - S(\rho) = \log(7/2). \tag{5}$$

In particular ρ is zero-mean. The largest remaining modulus in (4) is $\cos(\pi/7)$, so the actual factor is $\lambda = \cos^2(\pi/7) \in (0, 1)$.

For any numerical replacement $0 \le \ell \le 1$ of λ, put $A = 1 + \frac{5}{2}\ell$. Then $1 \le A \le 7/2$ and $B = \log 7 - \log A \ge \log 2 > 0$. At $C = \sqrt{2}$, the proposed right-hand side is bounded by

$$\frac{\log 7}{B}\,\frac{\log 2}{2} \le \frac{1}{2}\log 7 < \log 7 - \log 2.$$

The final strict inequality is equivalent to $7 > 4$. This contradicts (5) and proves the theorem. □

Corollary 2. *For every integer $n \ge 1$, the state $\rho_n = \rho^{\otimes n}$ in $(\mathbb{C}^7)^{\otimes n}$ violates* (BGJ 58) *at $C = 2^{n/2}$.*

Proof. Characteristic functions and convolution outputs factor under tensor products. A product of numbers of modulus at most one has modulus one only when every factor does. Thus $\mathcal{M}(\rho_n) = I_{7^n}/7^n$, and the largest nonunit modulus remains $\cos(\pi/7)$: attain it in one coordinate and use the origin in the others. Hence

$$R_n = 7^n, \quad \operatorname{Tr}\rho_n^2 = 2^{-n}, \quad S(\rho_n) = n\log 2, \quad D_n = n\log(7/2), \quad \log\delta_q[\rho_n] = \tfrac{n}{2}\log 2.$$

With $A_n = 1 + \lambda((7/2)^n - 1)$, the denominator $B_n = n\log 7 - \log A_n$ is at least $n\log 2$ for every $\lambda \in [0, 1]$. The proposed bound is therefore at most $\frac{n}{2}\log 7 < D_n$. □

3 The companion statements

Remark 45 also records the weaker estimate

$$D(\rho\|\mathcal{M}(\rho)) \;\le\; \frac{\log R(\rho) - S(\rho)}{\log R(\rho) - \log\bigl[1 + \lambda(R(\rho)\operatorname{Tr}\rho^2 - 1)\bigr] - S(\rho)}\log C, \tag{BGJ 59}$$

under a positive-denominator hypothesis.

Corollary 3. *For every ρ_n in Corollary 2, the denominator in* (BGJ 59) *is positive, and that inequality holds strictly for every $C \ge \delta_q[\rho_n]$.*

Proof. The actual factor satisfies $3/4 < \lambda = \cos^2(\pi/7) < 1$, since $0 < \pi/7 < \pi/6$. The notation from Corollary 2 gives

$$0 < B_n - n\log 2 = -\log(\lambda + (1 - \lambda)(2/7)^n) < -\log\lambda < \log(4/3) < \tfrac{1}{2}\log 2 \le \tfrac{n}{2}\log 2.$$

Here $\log(4/3) < \frac{1}{2}\log 2$ follows by squaring from $16/9 < 2$. The numerator in (BGJ 59) equals $D_n > 0$ and $\log C \ge \frac{n}{2}\log 2$. Its right-hand side is therefore strictly larger than D_n, proving the assertion without a numerical approximation. □

For completeness, the entropy-growth statement in the same remark holds for all computationally diagonal states on $(\mathbb{C}^7)^{\otimes n}$ with $s = t = 2$. Let X, Y be independent copies of any finitely supported random variable on $\mathbb{F}_7^n$. Then

$$H(2X + 2Y) \geq H(2X + 2Y \mid Y) = H(X).$$

Equality means that the conditional laws of $2X + 2Y$ are identical for all Y in its support. Fix a support point a. After dividing by two, this says that the law of X is invariant under every translation $y - a$ with y in the support. Let K be the subgroup generated by these differences. The support lies in $a + K$, and translation invariance forces the law to be uniform on that coset. Conversely, $2K = K$, so if X is uniform on $a + K$, then $2X + 2Y$ is uniform on $4a + K$ and has the same entropy. Such diagonal coset states are minimal stabilizer-projection states. Indeed, with $K^\perp$ the annihilator of K, the projection onto the coset's span is

$$P_{a+K} = \frac{1}{|K^\perp|} \sum_{p \in K^\perp} \omega^{-p \cdot a} Z^p, \qquad \rho_{a+K} = \frac{P_{a+K}}{|K|}.$$

Conversely, a diagonal minimal stabilizer-projection state has only clock coefficients in its Weyl expansion; its clock eigenvalue constraints define an affine subspace, and its normalized projection is uniform there. This proves the equality classification within the computationally diagonal class at these parameters, as mentioned in Remark 45, consistently with the classical zero-doubling characterization [4, Theorem 1.11(i), §5].

The counterexample family specifically violates (58), satisfies (59), and has strict entropy growth; the source's pure-state Theorem 44 is not refuted. The complementary quantum-difference constant of Remark 43 also equals $\sqrt{2}$ for the single-qudit example: the law of $-2I+2J$ has weights $1/2, 1/4, 1/4$ at $0, 2, 5$, respectively. A bound with the same right-hand side and that constant would fail for the same reason.

Verification. The proofs above cover every $n \geq 1$. An exact-arithmetic checker in the accompanying source archive verifies the seven-dimensional permutation, both convolution marginals, the characteristic-function modulus-one set in $\mathbb{Q}(e^{2\pi i/7})$, and tensor-output entropies for $1 \leq n \leq 8$. The archive's member manifest fixes the checker version; its finite checks corroborate, rather than establish, the all-n proofs.

References

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